Solution Method for the 3-ESP, 3-NSP Multi-Provider Extension

1 Overview

This document describes the solution method implemented in the multi-provider experiment scripts under `scripts/multi_provider`. The method extends the original ESP–NSP Stackelberg framework to a market with three edge service providers (ESPs), three network service providers (NSPs), and multiple users. It is intended as an experimental extension rather than a replacement of the main theoretical analysis in the paper. The implementation follows the same paper-facing idea: service providers set prices in Stage I, and users respond in Stage II through offloading, provider-pair selection, and resource purchase decisions.

Let

$$\mathcal{E} = \{1, 2, 3\}, \qquad \mathcal{N} = \{1, 2, 3\},$$

denote the ESP and NSP sets, respectively. Each ESP $e \in \mathcal{E}$ owns computation capacity F_e, unit cost c_e^E, and price p_e^E. Each NSP $n \in \mathcal{N}$ owns bandwidth capacity B_n, unit cost c_n^N, and price p_n^N. Each user can either compute locally or select one ESP–NSP pair (e, n) for offloading.

2 User Cost Model

The implementation stores the user-side constants

$$a_i \triangleq \alpha_i w_i, \qquad \theta_{i,n} \triangleq \frac{d_i(\alpha_i + \beta_i \rho_i \varpi_i)}{\sigma_{i,n}},$$

where $\sigma_{i,n}$ is the spectral efficiency from user i to NSP n. The local computation cost is denoted by C_i^ℓ.

If user i offloads through pair (e, n), buys computation resource f_i, and buys bandwidth b_i, its offloading cost is

$$C^o_{i,e,n}(f_i, b_i) = \frac{a_i}{f_i} + \frac{\theta_{i,n}}{b_i} + p_e^E f_i + p_n^N b_i. \tag{1}$$

The binary assignment variables can be written as

$$x_{i,e,n} \in \{0, 1\}, \qquad \sum_{e \in \mathcal{E}} \sum_{n \in \mathcal{N}} x_{i,e,n} \le 1.$$

When $\sum_{e,n} x_{i,e,n} = 0$, user i computes locally.

3 Stage-II User Response

Given provider prices

$$\boldsymbol{p} = (p_1^E, p_2^E, p_3^E, p_1^N, p_2^N, p_3^N),$$

the ideal Stage-II problem is a social-cost minimization over user assignment and resource allocation:

$$
\begin{aligned}
\min_{\boldsymbol{x},\boldsymbol{f},\boldsymbol{b}} \quad & \sum_i \left(1 - \sum_{e,n} x_{i,e,n}\right) C_i^\ell + \sum_i \sum_{e,n} x_{i,e,n} C^o_{i,e,n}(f_i, b_i) \\
\text{s.t.} \quad & \sum_i \sum_n x_{i,e,n} f_i \le F_e, \qquad \forall e \in \mathcal{E}, \\
& \sum_i \sum_e x_{i,e,n} b_i \le B_n, \qquad \forall n \in \mathcal{N}, \\
& C^o_{i,e,n}(f_i, b_i) \le C_i^\ell, \qquad \forall i, e, n \text{ with } x_{i,e,n} = 1.
\end{aligned}
\tag{2}
$$

This mixed-integer problem is expensive because users must choose among nine provider pairs. The experiment therefore uses a greedy assignment heuristic with exact fixed-assignment resource response.

3.1 Fixed-Assignment Closed-Form Response

Consider a fixed assignment A, where $A_i = (e_i, n_i)$ for offloading users and $A_i = \emptyset$ for local users. Let

$$
\mathcal{I}_e(A) = \{i : A_i = (e, n) \text{ for some } n\}, \qquad \mathcal{I}_n(A) = \{i : A_i = (e, n) \text{ for some } e\}.
$$

For each ESP and NSP, define the congestion threshold prices

$$
\bar{p}_e^E(A) = \left(\frac{\sum_{i \in \mathcal{I}_e(A)} \sqrt{a_i}}{F_e}\right)^2, \qquad \bar{p}_n^N(A) = \left(\frac{\sum_{i \in \mathcal{I}_n(A)} \sqrt{\theta_{i,n}}}{B_n}\right)^2. \tag{3}
$$

The effective prices are

$$
\tilde{p}_e^E(A) = \max\{p_e^E, \bar{p}_e^E(A)\}, \qquad \tilde{p}_n^N(A) = \max\{p_n^N, \bar{p}_n^N(A)\}. \tag{4}
$$

For any offloading user assigned to pair (e, n), the fixed-assignment resource response is

$$
f_i^*(A) = \sqrt{\frac{a_i}{\tilde{p}_e^E(A)}}, \qquad b_i^*(A) = \sqrt{\frac{\theta_{i,n}}{\tilde{p}_n^N(A)}}. \tag{5}
$$

The corresponding capacity shadow prices are

$$
\lambda_e(A) = \tilde{p}_e^E(A) - p_e^E, \qquad \nu_n(A) = \tilde{p}_n^N(A) - p_n^N. \tag{6}
$$

The implementation uses these expressions in `fixed_assignment_response`. They also make it cheap to evaluate the social cost and individual-rationality margin of an assignment:

$$
m_i(A) = C_i^\ell - C^o_{i,e_i,n_i}\big(f_i^*(A), b_i^*(A)\big). \tag{7}
$$

An assigned user is feasible only when $m_i(A) \ge 0$.

3.2 Greedy Admission and Rollback

Starting from the empty offloading assignment, the Stage-II heuristic admits users one by one. Given the current assignment A, each currently local user j evaluates every ESP–NSP pair using the congestion-adjusted score

$$h_{j,e,n}(A) = \min_{0<f\leq F_e} \left\{ \frac{a_j}{f} + (p_e^E + \lambda_e(A))f \right\} + \min_{0<b\leq B_n} \left\{ \frac{\theta_{j,n}}{b} + (p_n^N + \nu_n(A))b \right\} - C_j^\ell. \tag{8}$$

Each one-dimensional minimization has the bounded closed form

$$\min_{0<z\leq U} \left\{ \frac{\xi}{z} + \tau z \right\} = \begin{cases} 2\sqrt{\xi\tau}, & \sqrt{\xi/\tau} \leq U, \\ \xi/U + \tau U, & \sqrt{\xi/\tau} > U. \end{cases} \tag{9}$$

The algorithm selects the user-pair triple with the most negative score:

$$(j^*, e^*, n^*) \in \arg\min_{j,e,n} h_{j,e,n}(A).$$

If the minimum score is nonnegative, the algorithm stops. Otherwise, user j^* is tentatively assigned to (e^*, n^*), the fixed-assignment response is recomputed, and a rollback check is applied. The tentative admission is removed if it makes the realized social cost worse or if any offloading user violates individual rationality. Users that fail the rollback check are removed from the active candidate pool for that Stage-II solve.

The Stage-II output contains the final assignment, resources, shadow prices, social cost, number of accepted admissions, number of rollbacks, and the social cost trace.

4 Provider Revenues

After the Stage-II response is computed, each provider's realized demand and revenue are evaluated from the assigned users:

$$U_e^E(\boldsymbol{p}) = (p_e^E - c_e^E) \sum_{i\in\mathcal{I}_e(A^*(\boldsymbol{p}))} f_i^*, \qquad e \in \mathcal{E}, \tag{10}$$

$$U_n^N(\boldsymbol{p}) = (p_n^N - c_n^N) \sum_{i\in\mathcal{I}_n(A^*(\boldsymbol{p}))} b_i^*, \qquad n \in \mathcal{N}. \tag{11}$$

Here $A^*(\boldsymbol{p})$ denotes the assignment returned by the Stage-II heuristic at prices $\boldsymbol{p}$.

5 Stage-I Six-Provider Pricing Dynamics

The Stage-I game has six leaders: three ESPs and three NSPs. Exhaustive six-dimensional price search is not practical. The implementation therefore uses a boundary-price best-response heuristic inspired by the switching-boundary logic of the main paper.

5.1 Local Candidate Assignment Family

At a current price vector $\boldsymbol{p}$, the solver first computes the Stage-II response $A^*(\boldsymbol{p})$. It then constructs a local family of candidate assignments. Current offloading users are sorted by increasing individual-rationality margin m_i, so the first users are the easiest to remove. Current local users are sorted by their best pair score $\min_{e,n} h_{i,e,n}$, so the first users are the most likely to enter.

For a small integer Q, candidate assignments are created by removing the first r fragile offloading users and adding the first s promising local users, where

$$0 \le r \le Q, \qquad 0 \le s \le Q.$$

Duplicate and empty assignments are discarded. Hence the candidate family is small, with size at most $(Q+1)^2$ before duplicate removal.

5.2 Boundary Price for One Provider

For a provider k, all other providers' prices are fixed. For each local candidate assignment A, the solver estimates the right boundary of the assignment's feasible price interval for provider k. This boundary is the first price at which one user served by provider k becomes indifferent between offloading and local computation:

$$\min_{i \in \mathcal{I}_k(A)} m_i(A; p_k, p_{-k}) = 0. \tag{12}$$

In the implementation this boundary is computed by bracketing and bisection. If the candidate assignment is already infeasible at the provider's cost floor, it is skipped. If no zero-margin crossing is found before the configured maximum price, the maximum reachable price is used as the candidate boundary.

For each provider k, the evaluated price set contains:

- the current price p_k;
- all valid boundary prices from the local assignment family;
- one small upward exploratory price, $\min\{p_k^{\max}, \max(1.1p_k, p_k + 0.03)\}$.

Each price in this finite set is evaluated by rerunning the Stage-II heuristic under the corresponding unilateral deviation. The provider's estimated best response is the evaluated price with the largest realized revenue:

$$\widehat{p}_k^{\mathrm{BR}} \in \arg \max_{q_k \in \widehat{\mathcal{C}}_k(\boldsymbol{p})} U_k(q_k, p_{-k}). \tag{13}$$

The restricted best-response gain is

$$\widehat{G}_k(\boldsymbol{p}) = U_k(\widehat{p}_k^{\mathrm{BR}}, p_{-k}) - U_k(p_k, p_{-k}). \tag{14}$$

5.3 Iterative Pricing Rule

The restricted Nash-equilibrium gap is defined as

$$\widehat{\delta}(\boldsymbol{p}) = \max_{k \in \mathcal{E} \cup \mathcal{N}} \widehat{G}_k(\boldsymbol{p}). \tag{15}$$

At each Stage-I iteration, all six providers compute their estimated best-response gains. If

$$\widehat{\delta}(\boldsymbol{p}) \le \varepsilon,$$

the pricing dynamics terminates. Otherwise, the implementation updates the single provider with the largest restricted gain:

$$k^* \in \arg\max_k \widehat{G}_k(\boldsymbol{p}), \qquad p_{k^*} \leftarrow \widehat{p}_{k^*}^{\mathrm{BR}}.$$

All other provider prices remain unchanged in that iteration. After the final Stage-I iteration, the solver runs Stage II one more time and reports the final assignment, provider revenues, restricted gap, and resource utilization.

6 Algorithm Summary

Algorithm 1 Multi-Provider Stage-II Heuristic

Require: Prices $\boldsymbol{p}$, capacities $\{F_e\}$, $\{B_n\}$

Initialize empty assignment A and active local-user pool.
repeat
    Compute fixed-assignment response by (4)–(5).
    Roll back the last admitted user if the realized social cost does not decrease.
    Remove any assigned user violating $m_i(A) \geq 0$.
    For every active local user and every pair (e, n), compute $h_{i,e,n}(A)$.
    if the minimum score is negative **then**
        Tentatively assign the corresponding user to the corresponding pair.
    else
        Stop.
    end if
until the active pool is empty or the iteration limit is reached
Recompute the final fixed-assignment response and remove infeasible users.
return assignment, resources, shadow prices, social cost, and diagnostics.

Algorithm 2 Six-Provider Boundary-Price Pricing Dynamics

Require: Initial prices $\boldsymbol{p}^{(0)}$, local family size Q, tolerance ε

for $t = 0, 1, \ldots, T_{\max} - 1$ **do**
    Run Stage II at $\boldsymbol{p}^{(t)}$.
    for all providers $k \in \mathcal{E} \cup \mathcal{N}$ **do**
        Construct the local candidate assignment family.
        Compute candidate boundary prices using (12).
        Evaluate each candidate unilateral price by rerunning Stage II.
        Obtain $\widehat{p}_k^{\mathrm{BR}}$ and $\widehat{G}_k$.
    end for
    Set $\widehat{\delta}(\boldsymbol{p}^{(t)}) = \max_k \widehat{G}_k$.
    if $\widehat{\delta}(\boldsymbol{p}^{(t)}) \leq \varepsilon$ **then**
        Stop.
    end if
    Update only the provider $k^* \in \arg\max_k \widehat{G}_k$.
end for
Run Stage II at the final price vector.
return final prices, assignment, revenues, utilization, and diagnostics.

7 Implementation Parameters Used in the Reported Runs

The reported $N = 30$ and $N = 50$ runs use the `code_default_heterogeneous` setup, which follows the numerical scale of the experiment code rather than the physical-unit scale in the paper text. The user distributions are

$$\begin{array}{lll} w_i \sim U[0.5, 2.5], & d_i \sim U[1.0, 10.0], & f_i^\ell \sim U[0.5, 1.2], \\ \alpha_i \sim U[10.0, 15.0], & \beta_i \sim U[0.1, 0.5], & \rho_i \sim U[0.5, 2.0], \\ \varpi_i \sim U[0.8, 1.2], & \kappa_i \sim U[0.01, 0.05], & \sigma_{i,n}^{\text{base}} \sim U[0.5, 3.0]. \end{array}$$

The NSP channel qualities use the bias vector

$$[1.15,\ 1.00,\ 0.85],$$

clipped to the interval $[0.5, 3.0]$. The provider parameters are

$$(F_1, F_2, F_3) = (45, 35, 25), \qquad (B_1, B_2, B_3) = (20, 14, 10),$$

$$(c_1^E, c_2^E, c_3^E) = (0.008, 0.010, 0.013), \qquad (c_1^N, c_2^N, c_3^N) = (0.007, 0.010, 0.014).$$

The default initial prices are

$$\boldsymbol{p}_E^{(0)} = (0.42, 0.50, 0.58), \qquad \boldsymbol{p}_N^{(0)} = (0.40, 0.50, 0.62).$$

The $N = 30$ and $N = 50$ runs used $Q = 2$, maximum price 6.0 for both resource types, ten Stage-I iterations, and ninety-six Stage-II iterations per Stage-II solve.

8 Remarks

The method is intentionally heuristic. It preserves the main paper's boundary-price intuition, but the full 3-ESP, 3-NSP setting has a six-dimensional leader game and a nine-choice assignment problem on the user side. Therefore, the computed outcome should be interpreted as an approximate multi-provider Stackelberg outcome produced by a boundary-price restricted best-response dynamics. The diagnostics saved by the scripts, including assignment heatmaps, restricted-gap trajectories, provider revenues, and resource utilizations, are used to judge whether the resulting outcome is economically meaningful.